Saleh Mehdikhani

Embedded Software Engineer

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SUMMARY

I am a senior System Software Developer with 15+ years of experience in embedded systems and IoT. I build reliable, efficient, and maintainable firmware and system software, combining hands-on engineering with a product-minded approach. Throughout my career, I have delivered solutions from prototype to production, contributing both technically and strategically to the success of multiple products.

SKILLS

Programming Languages

C | C++ | Python | Go

Embedded OS

RTOS (Zephyr | FreeRTOS) | Embedded Linux | Docker

Embedded Systems

Low-Power Design | Real-Time Systems | Edge AI

Protocols

MQTT | RESTful API | CBOR | BLE | WiFi | UART | SPI | I²C | CAN | GPIO

Platforms

Silabs | Nordic | onsemi | TI | ST | Espressif | Atmel

Software Tools

Git | Github CI | CMake | Bazel | SCons | GNU Make

Hardware Tools

Logic Analyzer | Oscilloscope | Multimeter | Soldering | Power Analyzer

WORK EXPERIENCE (4)

Sep 2020 - Current

Embedded System Developer at Unikie

Finland

Working as a consultant on embedded and IoT projects for global customers across industries, from Finland to the US. Assigned to diverse client projects to complete critical development phases, ensure delivery quality, and integrate seamlessly with existing teams. This role demands rapid learning, adaptability, and a proactive approach to new technologies and environments.

May 2015 - May 2019

Product Manager / Embedded System Leader / Developer at HooshRavan

Tehran Province, Iran

Oversaw the complete lifecycle of multiple IoT products, from concept to production. Managed embedded and application teams, ensuring alignment between technology, product design, and manufacturing.

Dec 2011 - Mar 2016

Embedded System Developer at Hamayeh

Tehran, Iran

Developed embedded systems and PCBs for automation and lighting products. Improved firmware reliability and optimized development processes for faster delivery.

Oct 2009 - Mar 2012

Founder at Sepydar

Tehran Province, Iran

Founded a startup focused on modernizing industrial control systems. Designed and deployed controllers that remain operational in industrial environments.

PROJECTS (8)

Electrical Automotive Firmware

C | Bazel | Scons

Contributed to the development of an electric vehicle software platform, focusing on build system integration using Bazel and SCons. Worked in an agile, cross-cultural team with a U.S. customer to maintain multiple coexisting build systems, resolve dependency conflicts, and ensure continuous compatibility with a fast-evolving upstream codebase. Also supported new feature development and build optimization tasks.

AoA Firmware Development

https://www.digikey.fi/en/products/detail/corehw-semiconductor-ltd/CHW-LOC4000-1-0/25956605

C | BLE | Embedded Linux | MQTT

Led the firmware team in developing CoreHW's next-generation indoor positioning system based on a proprietary BLE protocol. Enhanced a sample system to support multi-device full-duplex communication while relaying IQ samples to MQTT in real time. Performed system-level analysis to identify and resolve bottlenecks, optimizing capacity and reliability. Actively contributed to overall system design and collaborated closely with a medium-sized cross-functional team to deliver a robust and high-performance firmware solution.

View on DigiKey | Demo

AI/ML TPMS

https://www.unikie.com/stories/unikie-silicon-labs-creating-on-device-ai-solutions/

C | Edge AI (TensorFlow Lite) | BLE

Developed an edge ML-based BLE direction-finding system running entirely on the MCU. Tuned AoA signal sampling, trained a machine learning model, and deployed it for on-device direction detection. Overcame signal noise challenges with a post-processing function that aggregates recent AI outputs for accurate estimation. Results were successfully demonstrated at Embedded World 2025.

View on Unikie | Demo

AoA SDK Sample

https://www.onsemi.com/company/news-media/press-announcements/en/onsemi-launches-end-to-end-positioning-system-to-enable-accurate-more-power-efficient-asset-tracking

C | CMSIS Pack | BLE

Developed firmware for BLE AoA positioning on RSL15 microcontrollers, implemented in Embedded C on top of the CMSIS Pack SDK. Delivered a complete AoA sample for both transmitter and receiver, which was published as the official Onsemi SDK example. Contributed to multi-sync support, resource management without an OS, and overall system design in collaboration with the customer. Overcame challenges in compatibility with SDK examples and constrained MCU resources to ensure robust and standards-compliant functionality.

View on onsemi | Demo

Zephyr AoA Firmware

C Zephyr BLE Multi-Threaded

Designed and implemented firmware for nRF52 series microcontrollers to enable BLE 5 AoA positioning, building on Zephyr OS. Developing a multi-device operation and maximized IQ sample throughput, overcoming early Zephyr limitations for AoA on nRF52833. Contributed fixes upstream to the Zephyr project and led the software architecture design, including multi-threading management, resource optimization, and reliable signal handling. This firmware formed the second generation of CoreHW's publicly available AoA system.

Security Platform

C CMake OpTEE

Developed and integrated a security platform based on OpTee for a proprietary product. Focused on resolving integration issues, and producing clean, maintainable CMake build scripts. Worked closely in a small team to ensure a robust and reliable secure firmware foundation.

Smart Home Devices

https://www.hinava.com/

C C++ BLE WiFi ZigBee

Served as the technical leader and embedded developer for a commercial smart home system. Designed the system architecture and developed devices using Zigbee (TI chipset), BLE (TI chipset), and WiFi (Espressif). Focused on mass-production readiness, optimizing for low power consumption**, cost efficiency, and real-time, stable communication across multiple protocols. Successfully delivered a commercial product now available in the market.
View on Hinava

Other Projects

C C++ Python

Worked on a range of projects spanning industrial automation, research, and telecommunications, including:

- Image Processing for Production Lines: Developed a Python + Qt/C++ system to detect missing components in real time for a manufacturing customer.
- Distributed Android Virtual Devices (Bosch Sub-company): Researched scalable deployment of Android applications in a cloud-based virtual environment.
- Baseband Bus Software (Nokia): Maintained and incrementally improved an existing baseband bus software solution, ensuring reliability and long-term support.

EDUCATION (2)

2011 - 2013

Master of Science (MS) Computer Architecture at Amirkabir University of Technology – Tehran Polytechnic

Introducing a Method to Distribute Processes on a Network of Smart Phones (Grade: A).

2007 - 2011

Bachelor of Science (BS) Computer Hardware Engineering at Iran University of Science and Technology (IUST)

Design of a Board Capable of Running a Customized Linux OS.